

#1. $R = 6578 \text{ km}$ $m = 100 \text{ kg}$. $\Rightarrow W = \sqrt{\mu/R^3} \approx 1.1834 \times 10^{-3} \text{ rad/s}$.

a. The approximation eg. $\begin{cases} m \dot{r} = F_r \Rightarrow m \ddot{r} = F_r \\ r m \dot{\Delta} = F_\Delta \Rightarrow m \ddot{\Delta} = F_\Delta \\ m r \dot{\Delta} = F_\lambda \Rightarrow m \ddot{\Delta} = F_\lambda \end{cases}$

where $x = r$, $y = r \Delta$, $z = r \lambda$.

관측이 없을 때 관측자의 해는,

$$x(t) = \dot{x}_0(t) + x_0, y(t) = \dot{y}_0(t) + y_0, z(t) = \dot{z}_0(t) + z_0.$$

where $x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0$ are the initial conditions.

원래의 CW의 해는 STM (state transition matrix)를 얻을 수 있는데 이를 선형화하여 근사하면 관측자의 해가 된다.

따라서 CW의 해에 있는 nonlinear term이 accuracy를 결정한다.

CW의 해에 있는 비선형 항은 $\cos(\omega t)$ 과 $\sin(\omega t)$ 이다.

$$\left| \frac{\omega t - \sin(\omega t)}{\sin \omega t} \right| \leq 0.05 \quad \& \quad \left| \frac{1 - \cos(\omega t)}{\cos \omega t} \right| \leq 0.05$$

$$\Rightarrow \begin{cases} -0.05 |\sin \omega t| \leq \omega t - \sin \omega t \leq 0.05 |\sin \omega t| \\ -0.05 |\cos \omega t| \leq 1 - \cos \omega t \leq 0.05 |\cos \omega t| \end{cases}$$

$\omega t \leq \pi/2$ 라 가정하자 (ωt 가 충분히 작아야 근사를 할 수 있으므로 타당한 가정이다).

$$\Rightarrow \begin{cases} \omega t - \sin \omega t \leq 0.05 \sin \omega t \dots (B) \\ 1 - \cos \omega t \leq 0.05 \cos \omega t \dots (A) \end{cases}$$

(A)를 풀면, $(1.05)^{-1} \leq \cos \omega t \Rightarrow \omega t \leq \cos^{-1}(1.05^{-1})$

$$\Rightarrow \omega t \leq 0.3098$$

$\omega t = 0.3098$ 을 (B)에 대입하면,

$$0.3098 - 1.05 \sin(0.3098) = -0.0103 < 0.$$

따라서 0.3098은 식 (A), (B)를 모두 만족하는 수 중 가장 큰 수이다.

$$\Rightarrow t \leq 0.3098 / W \approx 261.8277 \text{ sec} \approx 4.3638 \text{ min}.$$

7% 정확도를 가지도록 하는 time period T 는 아래의 식으로 얻을 수 있다.

$$T = \frac{2\pi}{W} \approx 265.4737 \text{ sec}.$$

이는 앞에서 구한 시간 보다 크므로 사용하지 않을 것이다.

$\therefore$ The maneuver time period $T = 261.8277 \text{ sec}$.

b. 관측은 x, y, z 가 모두 분리되어 독립적이다. 초기 조건에서 x 만 0이 아니므로 y 와 z 는 고려하지 않을 것이다.

$$x(t) = \dot{x}_0 t - 10. \Rightarrow 0 = x(60\text{s}) = \dot{x}_0 \cdot 60 - 10$$

$$\Rightarrow \dot{x}_0 = 1/6.$$

$$\dot{x}(t) = \dot{x}_0 = \dot{x}_F \Rightarrow \dot{x}_F = 1/6.$$

$$\therefore \Delta V_p = 1/6 \text{ m/s}$$

$$\Delta V_F = -1/6 \text{ m/s}$$

오메가가 영인가보다!!!
->아니야

#2. $\vec{r} = -445645.6\hat{i} + 500\hat{j} + 200\hat{k}$ km. $T = 17200$ s

$\dot{\vec{r}} = -0.2\hat{i} - 0.3\hat{j} + 0.4\hat{k}$ km/s

By Newton's Law, $\ddot{\vec{r}} = -\frac{\mu_E}{r_E^3} \vec{r}_E - \frac{\mu_M}{r_M^3} \vec{r}_M + \frac{\vec{F}}{m}$

where $r_E = \sqrt{(x - D_E)^2 + y^2 + z^2}$; $r_M = \sqrt{(x + D_M)^2 + y^2 + z^2}$

From $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, by Coriolis thm,

$\dot{\vec{r}} = \dot{x}\hat{i} + \dot{y}\hat{j} + \dot{z}\hat{k} + \vec{\omega} \times (x\hat{i} + y\hat{j} + z\hat{k})$

$= (\dot{x} - \omega y)\hat{i} + (\dot{y} + \omega x)\hat{j} + \dot{z}\hat{k}$

$\ddot{\vec{r}} = (\ddot{x} - \omega\dot{y})\hat{i} + (\ddot{y} + \omega\dot{x})\hat{j} + \ddot{z}\hat{k} + \dot{\vec{\omega}} \times (x\hat{i} - \omega y)\hat{i} + (\dot{y} + \omega x)\hat{j} + \dot{z}\hat{k}$

$= (\ddot{x} - 2\omega\dot{y} - \omega^2 x)\hat{i} + (\ddot{y} + 2\omega\dot{x} - \omega^2 y)\hat{j} + \ddot{z}\hat{k}$

$$\begin{cases} \ddot{x} - 2\omega\dot{y} - \omega^2 x = -\frac{\mu_E}{r_E^3} (x - D_E) - \frac{\mu_M}{r_M^3} (x + D_M) + \frac{F_x}{m} \\ \ddot{y} + 2\omega\dot{x} - \omega^2 y = -\frac{\mu_E}{r_E^3} y - \frac{\mu_M}{r_M^3} y + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E}{r_E^3} z - \frac{\mu_M}{r_M^3} z + \frac{F_z}{m} \end{cases}$$

Which is the nonlinear EoM for restric 3 body problem.

Let (x_L, y_L, z_L) be the position of L_2 point respect to the $\hat{i}\hat{j}\hat{k}$ frame.

$\Rightarrow \dot{x}_L = \dot{y}_L = \dot{z}_L = \ddot{x}_L = \ddot{y}_L = \ddot{z}_L = 0 = y_L = z_L$

Let $x = x_L + \Delta x$, $y = y_L + \Delta y = \Delta y$, $z = z_L + \Delta z = \Delta z$.

$$\frac{\mu_E}{r_E^3} = \frac{\mu_E}{[(x_L + \Delta x - D_E)^2 + \Delta y^2 + \Delta z^2]^{3/2}}$$

$$\approx \left(\frac{1}{(x_L - D_E)^2 + 2\Delta x(x_L - D_E)} \right)^{3/2} \mu_E$$

$$= \frac{\mu_E}{|x_L - D_E|^3} \left(1 + 2 \frac{\Delta x}{x_L - D_E} \right)^{-3/2} \approx \frac{\mu_E}{|x_L - D_E|^3} \left(1 - 3 \frac{\Delta x}{x_L - D_E} \right)$$

$$\frac{\mu_M}{r_M^3} = \frac{\mu_M}{[(x_L + \Delta x + D_M)^2 + \Delta y^2 + \Delta z^2]^{3/2}}$$

$$\approx \mu_M \left((x_L + D_M)^2 + 2\Delta x(x_L + D_M) \right)^{-3/2}$$

$$= \frac{\mu_M}{|x_L + D_M|^3} \left(1 + \frac{2\Delta x}{x_L + D_M} \right)^{-3/2} \approx \frac{\mu_M}{|x_L + D_M|^3} \left(1 - 3 \frac{\Delta x}{x_L + D_M} \right)$$

(i). $\ddot{\Delta x} - 2\omega\dot{\Delta y} - \omega^2 \Delta x - \cancel{\omega^2 x_L} = -W_E^2 (x_L + \Delta x - D_E) \left(1 - 3 \frac{\Delta x}{x_L - D_E} \right)$

$$- W_M^2 (x_L + \Delta x + D_M) \left(1 - 3 \frac{\Delta x}{x_L + D_M} \right) + \frac{F_x}{m}$$

$$= -W_E^2 (x_L - D_E) - W_M^2 (x_L + D_M) + 2W_E^2 \Delta x + 2W_M^2 \Delta x + F_x/m$$

$$\Rightarrow \ddot{\Delta x} - 2\omega\dot{\Delta y} - \omega^2 (2W_E^2/\omega^2 + 2W_M^2/\omega^2 + 1) \Delta x = F_x/m$$

$$\begin{aligned} \textcircled{a} \ddot{\Delta y} + 2W\dot{\Delta x} - W^2\Delta y &= -W_E^2\Delta y \left(1 - 3\frac{\Delta x}{x_L - D_E}\right) - W_M^2\Delta y \left(1 - 3\frac{\Delta x}{x_L + D_M}\right) + \frac{F_y}{m} \\ &= -(W_E^2 + W_M^2)\Delta y + F_y/m. \end{aligned}$$

$$\begin{aligned} \textcircled{a} \ddot{\Delta z} &= -W_E^2\Delta z \left(1 - 3\frac{\Delta x}{x_L - D_E}\right) - W_M^2\Delta z \left(1 - 3\frac{\Delta x}{x_L + D_M}\right) + \frac{F_z}{m} \\ &= -(W_E^2 + W_M^2)\Delta z + F_z/m \\ \Rightarrow \ddot{\Delta z} + (W_E^2 + W_M^2)\Delta z &= F_z/m. \end{aligned}$$

Replace $\Delta x, \Delta y, \Delta z$ to x, y, z .

$$\begin{cases} \ddot{x} - 2W\dot{y} + C_1 W^2 x = F_x/m \\ \ddot{y} + 2W\dot{x} + C_2 W^2 y = F_y/m \\ \ddot{z} + C_3 W^2 z = F_z/m \end{cases}$$

where $\begin{cases} C_1 = -2(W_E^2 + W_M^2)/W^2 - 1 \approx -17.3809 \\ C_2 = (W_E^2 + W_M^2)/W^2 - 1 \approx 2.1904 \\ C_3 = (W_E^2 + W_M^2)/W^2 \approx 3.1904 \end{cases}$

The initial state $X_0 = (x_0, \dot{x}_0, y_0, \dot{y}_0, z_0, \dot{z}_0)$
 $= (-1000, -0.2, 500, -0.3, 200, 0.4)$

The terminal state $X_f = (0, 0, 0, 0, 0, 0)$

$$\textcircled{b} \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dot{y} \\ \ddot{y} \\ \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -C_1 W^2 & 0 & 0 & 2W \\ 0 & 0 & 0 & 1 \\ 0 & -2W & -C_2 W^2 & 0 \\ 0 & 1 & 0 & 0 \\ -C_3 W^2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & 0 \\ 0 & \frac{1}{m} \end{bmatrix} \begin{bmatrix} F_x \\ F_y \end{bmatrix}$$

$$\begin{bmatrix} \dot{z} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -C_3 W^2 & 0 \end{bmatrix} \begin{bmatrix} z \\ \dot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} F_z.$$

$$\Rightarrow \begin{cases} \dot{X}_{in} = A_{in} X_{in} + B_{in} U_{in} \\ \dot{X}_{out} = A_{out} X_{out} + B_{out} U_{out} \end{cases}$$

$$A_{in} B_{in} = \begin{bmatrix} \frac{1}{m} & 0 \\ 0 & \frac{2W}{m} \\ 0 & \frac{1}{m} \\ -\frac{2W}{m} & 0 \end{bmatrix} \quad A_{in}^2 B_{in} = \begin{bmatrix} 0 & \frac{2W}{m} \\ \frac{W^2}{m}(-C_1 - 4) & 0 \\ -\frac{2W}{m} & 0 \\ 0 & \frac{W^2}{m}(-4 - C_2) \end{bmatrix}$$

$$A_{in}^3 B_{in} = \begin{bmatrix} \frac{W^2}{m}(-C_1 - 4) & 0 \\ 0 & -\frac{2W^3}{m}(C_1 + C_2 + 4) \\ 0 & -\frac{W^2}{m}(C_2 + 4) \\ 2\frac{W^3}{m}(C_1 + 4 + C_2) & 0 \end{bmatrix}$$

$$\Rightarrow \text{rank} \begin{bmatrix} 0 & 0 & \frac{1}{m} & 0 & 0 & \frac{2W}{m} & -\frac{W^2}{m}(C_1 + 4) & 0 \\ \frac{1}{m} & 0 & 0 & \frac{2W}{m} & \frac{W^2}{m}(-C_1 - 4) & 0 & 0 & -\frac{2W^3}{m}(C_1 + C_2 + 4) \\ 0 & 0 & 0 & \frac{1}{m} & -\frac{2W}{m} & 0 & 0 & -\frac{W^2}{m}(C_2 + 4) \\ 0 & \frac{1}{m} & -\frac{2W}{m} & 0 & 0 & -\frac{W^2}{m}(C_2 + 4) & \frac{2W^3}{m}(C_1 + 4 + C_2) & 0 \end{bmatrix}$$

= 4.

In plane dynamics is controllable.

$$A_{out} B_{out} = \begin{bmatrix} \frac{1}{m} \\ 0 \end{bmatrix} \Rightarrow \text{rank} \begin{bmatrix} 0 & \frac{1}{m} \\ \frac{1}{m} & 0 \end{bmatrix} = 2.$$

Hence Out of plane is also controllable.

C. Consider eigenvalues of system matrix $A_{in} \Rightarrow A_{out}$.

In plane dynamics.

$$\pm \text{eigenvalues} : \pm 2.1587 \omega_0, \pm 1.8627 \omega_0 j$$

Since there exist positive real eigenvalue, the in plane dynamics is not stable.

Out of plane dynamics.

$$\pm \text{eigenvalues} : \pm 1.7862 \omega_0 j.$$

Since the eigenvalues are pure complex number, we cannot define out of plane dynamics is stable OR not.

$$\#3 \quad m = 100 \text{ kg}, \quad x_0 = (D_E - D_M) / 2 = -187699.5 \text{ km}.$$

$$\dot{x}_0 = \dot{y}_0 = \dot{y}_0 = \dot{z}_0 = \dot{z}_0 = 0.$$

$$a. \quad \vec{F} = 0 \Rightarrow F_x = F_y = F_z = 0.$$

If the spacecraft in equilibrium, equation $\star$ in #2 holds with zero acceleration.

$$r_E = r_M = D/2 = 192374 \text{ km} = d.$$

$$\textcircled{1} \quad -\omega^2 x = + \frac{\mu_E}{d^2} - \frac{\mu_M}{d^2} = \frac{\mu_E - \mu_M}{d^2}$$

$$\text{Left Side} : -\omega^2 x \approx 1.3301 \text{e-}06.$$

$$\text{Right Side} : (\mu_E - \mu_M) / d^2 = 1.0639 \text{e-}05.$$

Since the order of both sides are different, $\textcircled{1}$ does not hold for midway point.

$\textcircled{2}$ $\textcircled{3}$ hold for midway point because both sides are zero.

Hence the spacecraft is not in equilibrium.

The acceleration $\vec{a}$ is,

$$\ddot{x} = \omega^2 x + \frac{\mu_E - \mu_M}{d^2} \approx 9.3086 \text{e-}06 \text{ km/s}^2.$$

$$\Rightarrow \vec{a} = 9.3086 \text{e-}06 \hat{i} \text{ km/s}^2.$$

b. To keep the spacecraft in equilibrium, the propulsive should be same as $-\vec{a}$.

$$\Rightarrow \frac{\vec{F}}{m} = -\vec{a} = -9.3086 \text{e-}06 \hat{i} \text{ km/s}^2.$$